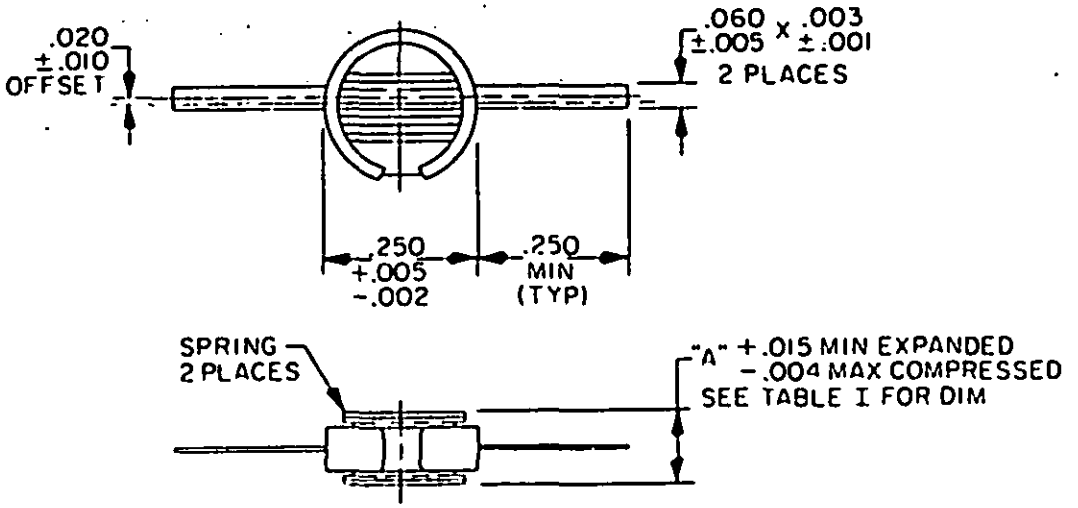
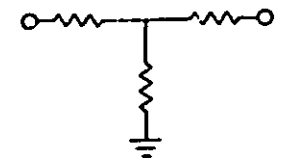
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, STRIPLINE, CLASS IV, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.





SCHEMATIC
BALANCED TO GROUND

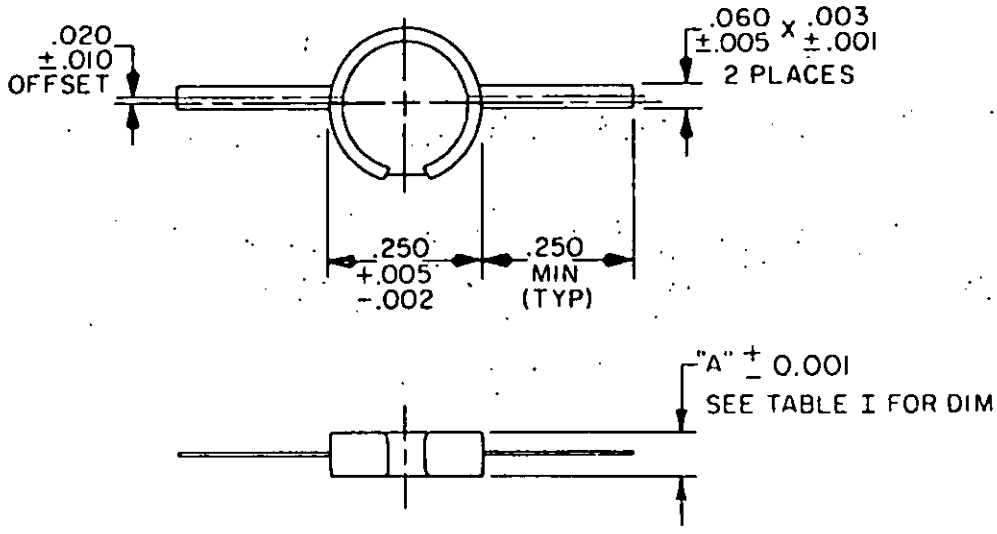
INCHES	MM	INCHES	MM
.001	.03	.010	.25
.002	.05	.015	.38
.003	.08	.020	.51
.004	.10	.060	1.52
.005	.13	.250	6.35

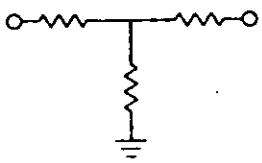
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Outline drawing and schematic for P/Ns M3933/22-01 through -11.

FSC 5985

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SCHEMATIC

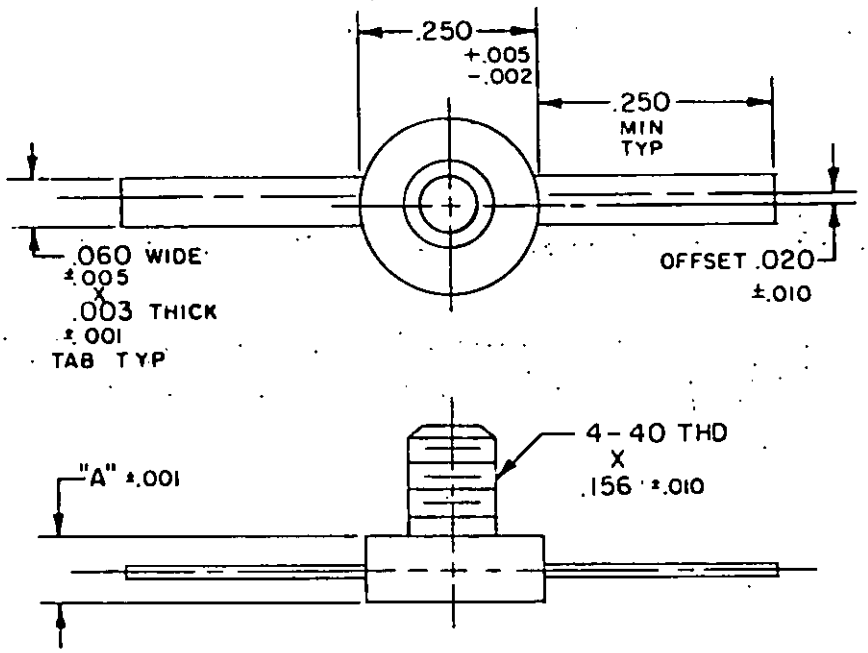
BALANCED TO GROUND

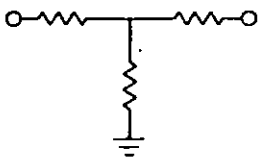
INCHES	MM	INCHES	MM
.001	.03	.010	.25
.002	.05	.020	.51
.003	.08	.060	1.52
.005	.13	.250	6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Outline drawing and schematic for P/Ns M3933/22-12 through -19.

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SCHEMATIC

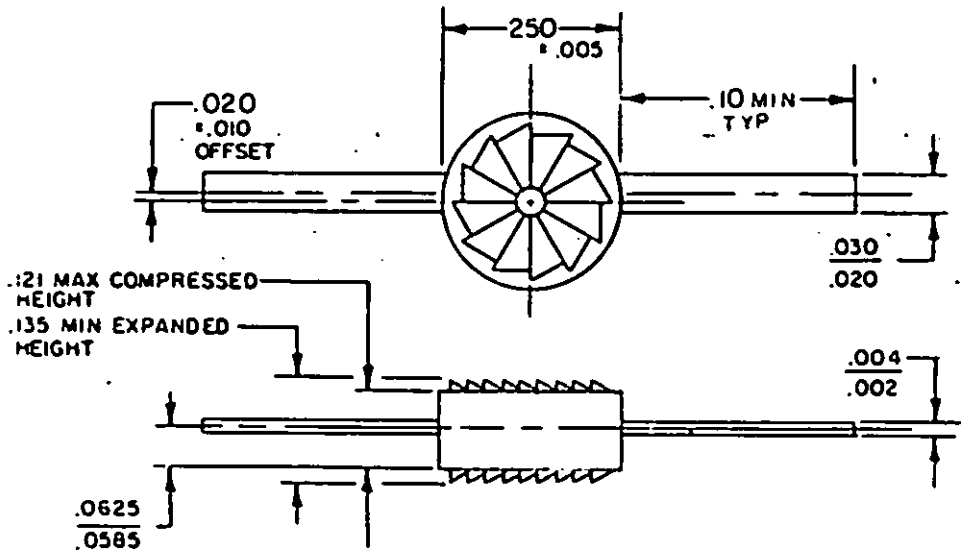
BALANCED TO GROUND

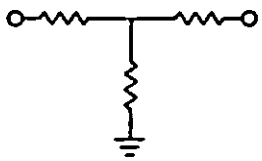
INCHES	MM	INCHES	MM
.001	.03	.010	.25
.002	.05	.015	.38
.003	.08	.020	.51
.004	.10	.060	1.52
.005	.13	.250	6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Outline drawing and schematic for P/Ns M3933/22-20 through -22.

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SCHEMATIC
BALANCED TO GROUND

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Outline drawing and schematic for P/Ns M3933/22-23 through -42.

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ENGINEERING DATA:

Class: IV.

Operating frequency range: DC - 18 gigahertz (GHz) (see table I).

Test frequencies for VSWR and attenuation measurements:

DC, 4, 8, and 12 GHz for dash numbers 01 thru 22.
DC, 4, 8, 12, and 18 GHz for dash numbers 23 thru 42.

Basic test frequency: 8 GHz.

Operating temperature range: -55°C to +125°C.

Storage temperature range: -62°C to 125°C.

Case temperature: +125°C.

Lead temperature: 0.062 inch from case for 10 seconds - +250°C.

Material: Case - Brass per QQ-B-626.

Tabs - Brass per QQ-B-613.

Contacts - Beryllium copper per QQ-C-533.

Lead finish: Plating per MIL-G-45204, type II, class 2.

Weight: 0.05 ounces nominal.

REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, 3 and 4 and table I.

Impedance: 50 ohms.

VSWR: See table I.

Attenuation: See table I.

Power: See table I.

Stability of attenuation:

After vibration: Test condition B (properly mounted in a stripline package).

After shock: Test condition G.

Part number: M3933/22- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB				Maximum power input @ 25°C		VSWR maximum				Fig.	2/ Dimension A (inches) maximum
	Nominal	Deviation			AV (W) 1/ (continuously)	Peak (W) 3/	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC	DC to 12.4 GHz	DC to 18 GHz								
01 N S	3	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
02 N S	6	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
03 N S	10	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
04 N S	20	±0.5	±1.5	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.250 (3.175)
05 N S	1	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
06 N S	5	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
07 N S	7	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
08 N S	8	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
09 N S	12	±0.5	±0.7	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
10 N S	15	±0.5	±1.0	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)
11 N S	17	±0.5	±1.2	---	1	50	1.30:1	1.40:1	1.50:1	---	1	0.125 (3.175)

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB				Maximum power input 0 25°C		VSWR maximum				Fig.	2/ Dimension A (inches) maximum
	Nominal	Deviation DC	Deviation DC to 12.4 GHz	Deviation DC to 18 GHz	AV (W) 1/ (continuously)	Peak (W) 3/	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
12 N S	1	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
13 N S	2	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
14 N S	3	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
15 N S	4	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
16 N S	6	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
17 N S	8	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
18 N S	10	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
19 N S	20	±0.5	±1.5	---	1	50	1.30:1	1.40:1	1.50:1	---	2	0.125 (3.175)
20 N S	3	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	3	0.125 (3.175)
21 N S	6	±0.2	±0.6	---	1	100	1.30:1	1.40:1	1.50:1	---	3	0.125 (3.175)
22 N S	10	±0.2	±0.6	---	1	50	1.30:1	1.40:1	1.50:1	---	3	0.125 (3.175)

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB				Maximum power input @ 25°C		VSWR maximum				Fig.	2/ Dimension A (inches) maximum
	Nominal	Deviation			AV (W) 1/ (continuously)	Peak (W) 3/	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC	DC to 12.4 GHz	DC to 18 GHz								
23 N S	1	±0.2	±0.5	±0.7	1	100	1.30:1	1.30:1	1.60:1	1.80:1	4	---
24 N S	2	±0.2	±0.5	±0.7	1	100	1.30:1	1.30:1	1.60:1	1.80:1	4	---
25 N S	3	±0.2	±0.5	±0.7	1	100	1.30:1	1.30:1	1.60:1	1.80:1	4	---
26 N S	4	±0.2	±0.6	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
27 N S	5	±0.2	±0.7	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
28 N S	6	±0.2	±0.7	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
29 N S	7	±0.2	±0.7	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
30 N S	8	±0.2	±0.7	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
31 N S	9	±0.2	±0.7	±0.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
32 N S	10	±0.2	±0.5	±0.5	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
33 N S	11	±0.2	±0.5	±0.5	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB				Maximum power input @ 25°C		VSWR maximum				Fig.	2/ Dimension A (inches) maximum
	Nominal	Deviation			AV (W) 1/ (continuously)	Peak (W) 3/	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC	DC to 12.4 GHz	DC to 18 GHz								
34 N S	12	±0.3	±0.5	±1.0	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
35 N S	13	±0.3	±0.8	±1.0	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
36 N S	14	±0.5	+0.0 -1.5	+0.0 -1.7	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
37 N S	15	±0.7	+0.0 -2.0	+0.0 -2.0	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
38 N S	16	±0.7	+0.0 -2.0	+0.0 -2.3	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
39 N S	17	±0.7	+0.0 -2.0	+0.0 -2.5	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
40 N S	18	±0.8	+0.0 -2.5	+0.0 -3.0	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
41 N S	19	±0.8	+0.0 -3.0	+0.0 -3.5	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---
42 N S	20	±1.0	+0.0 -3.3	+0.0 -3.5	1	100	1.30:1	1.40:1	1.60:1	1.80:1	4	---

1/ Power is derated linearly to .1 watts at +125°C.

2/. See outline drawing for tolerance.

3/ Peak power for a duty cycle of 1 percent, with a maximum pulse width of 10 microseconds.

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Note: Attenuators covered by this specification sheet are substitutable for the manufacturer's or user's part numbers specified in table III. This information in no way implies that manufacturer's part number is suitable as a substitute for the military part number.

TABLE II. Substitution information.

Military part number M3933/22-	Manufacturer's code	Manufacturer's part number	NSN
01	19202	11711156-003	---
01	24602	05-125-A-3	6625-00-004-1635
02	19202	11711156-001	---
02	24602	05-125-A-6	5985-01-065-9385
02	49956	912575	---
03	19202	11711156-002	---
03	24602	05-125-A-10	5985-01-031-3095
04	05869	712462-1	---
04	24602	05-250-A-20	---
05	49956	3141371-1	---
05	24602	05-125-A-1	---
06	24602	05-125-A-5	5985-01-111-5159
07	49956	3141371-3	---
07	24602	05-125-A-7	5985-01-111-5158
08	24602	05-125-A-8	5985-01-094-0957
09	24602	05-125-A-12	5985-01-094-0956
10	49956	3141371-4	---
10	24602	05-125-A-15	---
11	24602	05-125-A-17	5985-01-112-0009
12	24602	04-125-A-1	5905-01-075-3256
13	24602	04-125-A-2	5905-01-074-1782
14	24602	04-125-A-3	5985-01-073-1259
15	24602	04-125-A-4	5905-01-074-1781

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TABLE II. Substitution information - Continued.

Military part number M3933/22-	Manufacturer's code	Manufacturer's part number	NSN
16	24602	04-125-A-6	5905-01-086-2247
17	24602	04-125-A-8	5905-01-074-1780
18 18	24602 24602	04-125-A-10 04-125-A-10	5985-01-073-2431 5905-01-074-1779
19	24602	04-125-A-20	5905-01-074-1599
20	24602	09-125-A-3	5985-01-030-4068
21	24602	09-125-A-6	5985-00-715-5217
22	24602	09-125-A-10	5985-02-009-9052
23 23	99971 34677	7546669P1 0910-0108-1	--- ---
24 24	99971 34677	7546669P2 0910-0108-2	--- ---
25 25	99971 34677	7546669P3 0910-0108-3	--- ---
26 26	99971 34677	7546669P4 0910-0108-4	--- ---
27 27	99971 34677	7546669P5 0910-0109-5	--- ---
28 28	99971 34677	7546669P6 0910-0108-6	--- ---
29 29	99971 34677	7546669P7 0910-0108-7	--- ---
30 30	99971 34677	7546669P8 0910-0108-8	--- ---
31 31	99971 34677	7546669P9 0910-0108-9	--- ---

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TABLE II. Substitution information - Continued.

Military part number M3933/22-	Manufacturer's code	Manufacturer's part number	NSN
32	99971	7546669P10	---
32	34677	0910-0108-10	---
33	99971	7546669P11	---
33	34677	0910-0108-11	---
34	99971	7546669P12	---
34	34677	0910-0108-12	---
35	99971	7546669P13	---
35	34677	0910-0108-13	---
36	99971	7546669P14	---
36	34677	0910-0108-14	---
37	99971	7546669P15	---
37	34677	0910-0108-15	---
38	99971	7546669P16	---
38	34677	0910-0108-16	---
39	99971	7546669P17	---
39	34677	0910-0108-17	---
40	99971	7546669P18	---
40	34677	0910-0108-18	---
41	99971	7546669P19	---
41	34677	0910-0108-19	---
42	99971	7546669P20	---
42	34677	0910-0108-20	---

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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